Oxford College of Emory University

Chapter 14, Part 2

Money Creation and the Federal Reserve

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Roadmap

1. The Federal Reserve System
2. Loan creation, settlement, and repayment
3. A conditional deposit-expansion model
4. Practice and an individual checkpoint

The Federal Reserve System

Institution	Role
Board of Governors	Federal oversight and system responsibilities
12 Reserve Banks	Regional operations, financial services, and supervision
Federal Open Market Committee (FOMC)	Decisions about the monetary-policy stance

1913: the Federal Reserve Act established the Fed to address banking-system stresses.

Source: Federal Reserve Board, The Fed Explained.

Bank Accounting

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

Reserves and loans are assets. Deposits are liabilities. Equity is the residual.

Loan Origination

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

The bank credits a new $120 loan to a customer's deposit.

Loans: 900 + 120 = 1,020. Deposits: 1,000 + 120 = 1,120.

Reserves remain 200; equity remains 100.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Payment to Another Bank

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,120
Loans: 1,020	Equity: 100
Total: 1,220	Total: 1,220

The borrower pays $120 to a seller at Bank B.

Bank A: reserves −120; deposits −120.

Bank B: reserves +120; seller's deposits +120.

The Banking System

Change since before the loan	Bank A	Bank B	Combined
Reserves	−120	+120	0
Loans	+120	0	+120
Deposits	0	+120	+120

The payment moves deposits; the loan created the additional $120 of deposits.

Cash Withdrawals

- A customer withdraws $50: bank reserves −50 and deposits −50.
- Public currency rises by $50, offsetting the fall in deposits.
- Currency + deposits is unchanged; the bank has less liquidity.

Principal Repayment

- A borrower uses $50 in a deposit at the same bank to repay principal.
- Loans fall by $50 and deposits fall by $50.
- Currency + deposits falls by $50; reserves and equity do not change.

Loan Losses and Bank Capital

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

The bank writes off $80 of unrecoverable loans.

Loans fall to 820; equity falls to 20.

Deposits and reserves remain unchanged at the write-off.

Constraints on Lending

- **Capital**: losses and additional assets affect the equity cushion.
- **Liquidity and funding**: payments can leave the bank.
- **Risk and profitability**: borrowers must qualify, and lending must cover costs.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

A Reserve-Ratio Model

- Let banks hold reserves equal to **10% of deposits**: $r = 0.10$.
- All loan proceeds are redeposited; no extra reserves or currency leakage.
- Banks lend each available amount; borrowers want the loans; other constraints do not bind.

Successive Deposit Rounds

Round	New deposit ($)	Reserves retained	New loan
1	100.00	10.00	90.00
2	90.00	9.00	81.00
3	81.00	8.10	72.90

The central bank first buys a $100 security from a nonbank seller, adding reserves and a deposit.

The Deposit Multiplier

$$\Delta D = 100 + 90 + 81 + \cdots = \frac{100}{0.10} = 1{,}000$$

- **Total new deposits**: \$1,000, including the initial \$100.
- **Additional loans**: \$900. Total extra reserves: \$100.
- These are limiting model results under the stated conditions.

Depositing Existing Currency

A customer instead deposits $100 of currency already held outside banks.

Initial change: currency −100; deposits +100. No initial increase in currency + deposits.

With the same expansion conditions, deposits rise by 1,000 in total, but currency + deposits rises by 900.

The Model and U.S. Banking

- The U.S. reserve requirement has been **0% since March 26, 2020**.
- Banks still need liquidity and capital and must manage funding and credit risk.
- $1/r$ at $r = 0$ does not predict infinite lending or an infinite money supply.

Source: Federal Reserve Board, Reserve Requirements.

Central-Bank Security Purchases

Transaction	Reserves	Nonbank deposits
Buy from a bank	Bank reserves rise	No immediate required change
Buy from a nonbank seller	Seller’s bank reserves rise	Seller’s deposit rises

Both transactions exchange assets; the immediate effect depends on the seller.

Balance-Sheet Practice

Use the worksheet's bank: reserves 150, loans 750, securities 100, deposits 900, equity 100.

Record a new $90 loan, then its payment to another bank.

For each event, verify assets = liabilities + equity and state what happens to total deposits.

Bank Accounting Checkpoint

A bank writes off $30 of loans, with no repayment or deposit withdrawal. What changes immediately?

Loans −30 and equity −30.

Deposits and reserves do not change solely because of the write-off.

Banking and Monetary Policy

- Bank lending changes deposits; settlement moves reserves.
- The multiplier depends on its model conditions.
- Chapter 15 connects central-bank policy, interest rates, spending, and AD.